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Section: CSJ (Group 2)

Subject: NV62004

Assignment-1

1. Write a Dart program that accepts user input and displays it in a formatted output. Create a class in Dart for a Student with properties like name, rollNo, and marks. Add a method to display details.

```
import 'dart:io';

class Student {
  String name;
  int rollNo;
  double marks;

  Student(this.name, this.rollNo, this.marks);

  void displayDetails() {
    print("\n----- Student Details -----");
    print("Name : $name");
    print("Roll No: $rollNo");
    print("Marks : $marks");
  }
}

void main() {
  stdout.write("Enter student name: ");
  String name = stdin.readLineSync()!;
```

```

stdout.write("Enter roll number: ");

int rollNo = int.parse(stdin.readLineSync(!));

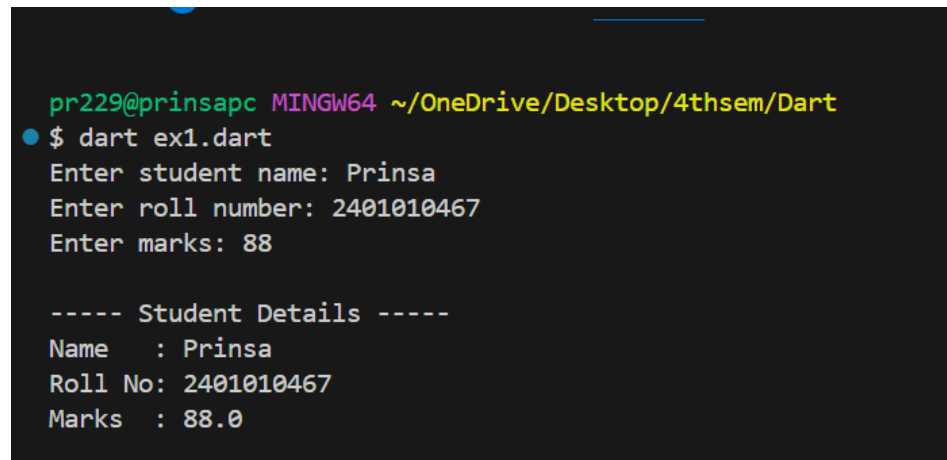
stdout.write("Enter marks: ");

double marks = double.parse(stdin.readLineSync(!));

Student s1 = Student(name, rollNo, marks);
s1.displayDetails();
}

```

Output:



```

pr229@prinsapc MINGW64 ~/OneDrive/Desktop/4thsem/Dart
$ dart ex1.dart
Enter student name: Prinsa
Enter roll number: 2401010467
Enter marks: 88

----- Student Details -----
Name    : Prinsa
Roll No: 2401010467
Marks   : 88.0

```

2. Implement inheritance in Dart for two classes: Person and Employee. What are constructors? Create a parameterized constructor for a Product class in Dart.

Employee inherits from Person:

```

class Person {
  String name;
  int age;

  Person(this.name, this.age);

  void displayPerson() {

```

```
    print("Name: $name");
    print("Age: $age");
  }
}

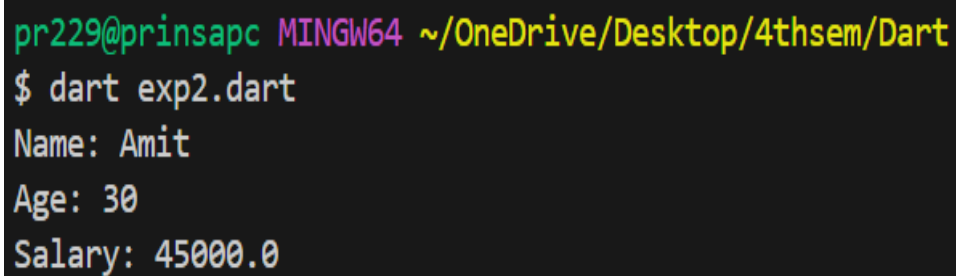
class Employee extends Person {
  double salary;

  Employee(String name, int age, this.salary) : super(name, age);

  void displayEmployee() {
    displayPerson();
    print("Salary: $salary");
  }
}

void main() {
  Employee emp = Employee("Amit", 30, 45000);
  emp.displayEmployee();
}
```

Output:

A screenshot of a terminal window with a dark background. The prompt is 'pr229@prinsapc' in green, followed by 'MINGW64' in pink and the path '~/OneDrive/Desktop/4thsem/Dart' in yellow. The command '\$ dart exp2.dart' is entered in white. The output is displayed in white text: 'Name: Amit', 'Age: 30', and 'Salary: 45000.0' on separate lines.

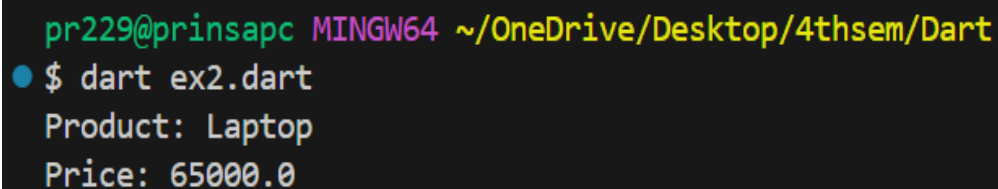
```
pr229@prinsapc MINGW64 ~/OneDrive/Desktop/4thsem/Dart
$ dart exp2.dart
Name: Amit
Age: 30
Salary: 45000.0
```

A constructor is a special method in a class used to initialize objects when they are created. It has the same name as the class and runs automatically when an object is created.

Parameterized Constructor (Product class):

```
class Product {  
    String name;  
    double price;  
    Product(this.name, this.price);  
  
    void displayProduct() {  
        print("Product: $name");  
        print("Price: $price");  
    }  
}  
  
void main() {  
    Product p1 = Product("Laptop", 65000);  
    p1.displayProduct();  
}
```

Output:

A screenshot of a terminal window with a dark background. The prompt is 'pr229@prinsapc MINGW64 ~/OneDrive/Desktop/4thsem/Dart'. The command '\$ dart ex2.dart' has been executed, resulting in two lines of output: 'Product: Laptop' and 'Price: 65000.0'.

```
pr229@prinsapc MINGW64 ~/OneDrive/Desktop/4thsem/Dart  
● $ dart ex2.dart  
Product: Laptop  
Price: 65000.0
```

3. Design a Flutter UI for a student registration form with Text Fields and buttons. Develop a Flutter app using Column and Row widgets to show a profile card layout.

Student Registration Form:

```
import 'package:flutter/material.dart';

void main() {
  runApp(const StudentFormApp());
}

class StudentFormApp extends StatefulWidget {
  const StudentFormApp({super.key});

  @override
  State<StudentFormApp> createState() => _StudentFormAppState();
}

class _StudentFormAppState extends State<StudentFormApp> {
  final nameController = TextEditingController();
  final deptController = TextEditingController();
  final semController = TextEditingController();

  void clearFields() {
    nameController.clear();
    deptController.clear();
    semController.clear();
  }

  void submitForm() {
    print("Name: ${nameController.text}");
```

```
print("Department: ${deptController.text}");  
print("Semester: ${semController.text}");  
}
```

```
@override
```

```
Widget build(BuildContext context) {  
  return MaterialApp(  
    home: Scaffold(  
      backgroundColor: Colors.grey[100], // subtle background  
      appBar: AppBar(  
        title: const Text("Student Registration"),  
        centerTitle: true,  
        backgroundColor: Colors.teal,  
      ),  
      body: Center(  
        child: Container(  
          width: 420,  
          padding: const EdgeInsets.all(25),  
          decoration: BoxDecoration(  
            color: Colors.white,  
            borderRadius: BorderRadius.circular(18),  
            boxShadow: const [  
              BoxShadow(color: Colors.black12, blurRadius: 12)  
            ],  
          ),  
          child: Column(  
            mainAxisAlignment: MainAxisAlignment.min,  
            children: [  
              Text("Student Registration"),  
              Text("Semester: ${semController.text}"),  
              Text("Department: ${deptController.text}"),  
            ],  
          ),  
        ),  
      ),  
    ),  
  );  
}
```

```
const Text(  
  "Student Form",  
  style: TextStyle(  
    fontSize: 22, fontWeight: FontWeight.bold),  
  ),  
const SizedBox(height: 20),
```

```
TextField(  
  controller: nameController,  
  decoration: const InputDecoration(  
    labelText: "Student Name",  
    border: OutlineInputBorder(),  
  ),  
,  
const SizedBox(height: 15),
```

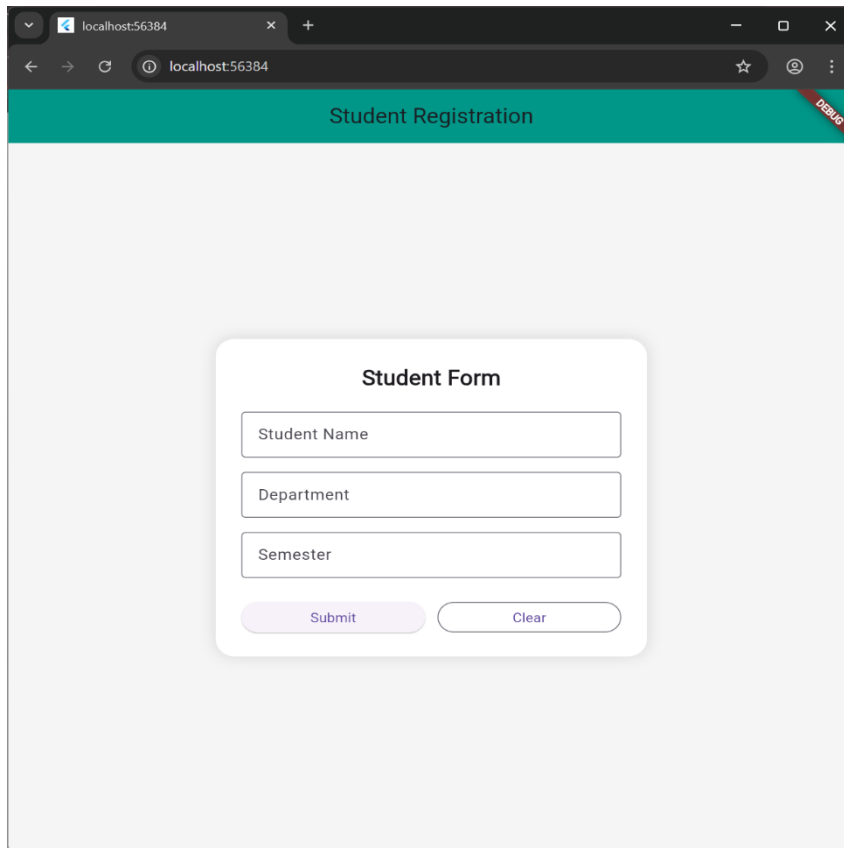
```
TextField(  
  controller: deptController,  
  decoration: const InputDecoration(  
    labelText: "Department",  
    border: OutlineInputBorder(),  
  ),  
,  
const SizedBox(height: 15),
```

```
TextField(  
  controller: semController,  
  decoration: const InputDecoration(  
    labelText: "Semester",  
    border: OutlineInputBorder(),  
  ),  
,  
const SizedBox(height: 15),
```



```
    ),  
  );  
}  
}
```

Output:



Profile Card Layout:

```
import 'package:flutter/material.dart';  
  
void main() {  
  runApp(const ProfileCardApp());  
}  
  
class ProfileCardApp extends StatelessWidget {  
  const ProfileCardApp({super.key});
```

```
@override

Widget build(BuildContext context) {

  return MaterialApp(

    home: Scaffold(

      backgroundColor: Colors.grey[100],

      appBar: AppBar(

        title: const Text("Student Profile"),

        centerTitle: true,

        backgroundColor: Colors.teal,

      ),

      body: Center(

        child: Container(

          width: 320,

          padding: const EdgeInsets.all(20),

          decoration: BoxDecoration(

            color: Colors.blue,

            borderRadius: BorderRadius.circular(25),

            boxShadow: const [

              BoxShadow(color: Colors.black26, blurRadius: 10),

            ],

          ),

          child: Row(

            children: [
```

```
const Icon(Icons.person, color: Colors.white, size: 50),

const SizedBox(width: 15),

Column(

  mainAxisAlignment: MainAxisAlignment.min,

  crossAxisAlignment: CrossAxisAlignment.start,

  children: const [

    Text(

      "Prinsa Tuladhar",

      style: TextStyle(

        color: Colors.white,

        fontSize: 20,

        fontWeight: FontWeight.bold,

      ),

    ),

    SizedBox(height: 6),

    Text(

      "Department: CSE",

      style: TextStyle(color: Colors.white),

    ),

    Text("Semester: 4", style: TextStyle(color: Colors.white)),

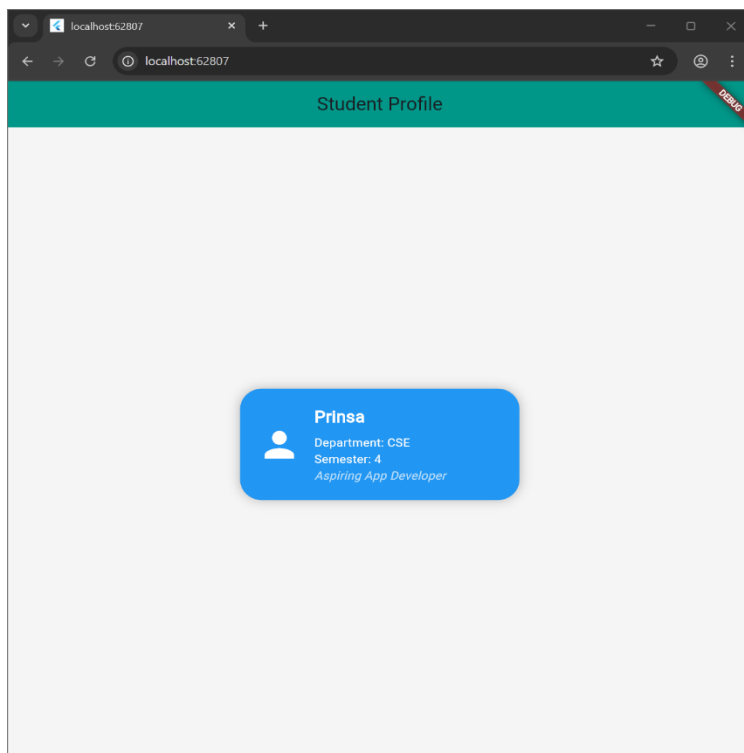
    Text(

      "Aspiring App Developer",

      style: TextStyle(
```

```
color: Colors.white70,  
fontStyle: FontStyle.italic,  
)  
)  
],  
)  
],  
)  
)  
)  
)  
)  
)  
)  
};  
} }
```

Ouput:



4. Develop a Flutter app using Column and Row widgets to show a profile card layout. Create a navigation from the Home Screen to the Details Screen and pass data between them.

```
import 'package:flutter/material.dart';

void main() {

  runApp(const MyApp());

}

class MyApp extends StatelessWidget {

  const MyApp({super.key});

  @override

  Widget build(BuildContext context) {

    return MaterialApp(home: const HomeScreen());

  }

}

// Home Screen

class HomeScreen extends StatelessWidget {

  const HomeScreen({super.key});

  @override

  Widget build(BuildContext context) {

    return Scaffold(

      appBar: AppBar(title: const Text('Home Screen')),

      body: Center(

        child: Column(

          mainAxisAlignment: MainAxisAlignment.center,
```

```
children: [

// Profile Card

Container(

width: 300,

padding: const EdgeInsets.all(16),

decoration: BoxDecoration(

color: Colors.white,

borderRadius: BorderRadius.circular(10),

boxShadow: const [

BoxShadow(

color: Colors.black12,

blurRadius: 5,

offset: Offset(0, 2),

),

],

),

child: Column(

crossAxisAlignment: CrossAxisAlignment.start,

children: [

// Row with Icon and Name

Row(

children: const [

Icon(Icons.person, color: Colors.purple, size: 40),
```

```
SizedBox(width: 10),

Column(

  crossAxisAlignment: CrossAxisAlignment.start,

  children: [

    Text(

      'Prinsa',

      style: TextStyle(

        fontSize: 18,

        fontWeight: FontWeight.bold,

      ),

    ),

    Text(

      'B.Tech CSE',

      style: TextStyle(fontSize: 14, color: Colors.grey),

    ),

  ],

),

const SizedBox(height: 16),

// View Details Button

Center(

  child: ElevatedButton(
```

```
onPressed: () {  
  Navigator.push(  
    context,  
    MaterialPageRoute(  
      builder: (context) => const DetailsScreen(  
        name: 'Prinsa ',  
        course: 'B.Tech CSE',  
      ),  
    ),  
  );  
},  
child: const Text('View Details'),  
),  
],  
),  
),  
],  
),  
),  
],  
),  
),  
];  
}  
}
```



```
// Details Screen

class DetailsScreen extends StatelessWidget {

  final String name;

  final String course;

  const DetailsScreen({super.key, required this.name, required this.course});

  @override

  Widget build(BuildContext context) {

    return Scaffold(

      appBar: AppBar(title: const Text('Details Screen')),

      body: Center(

        child: Column(

          mainAxisAlignment: MainAxisAlignment.center,

          children: [

            // Title

            const Text(

              'Student Details',

              style: TextStyle(fontSize: 24, fontWeight: FontWeight.bold),

            ),

            const SizedBox(height: 30),

            // Details Container

            Container(

              width: 300,

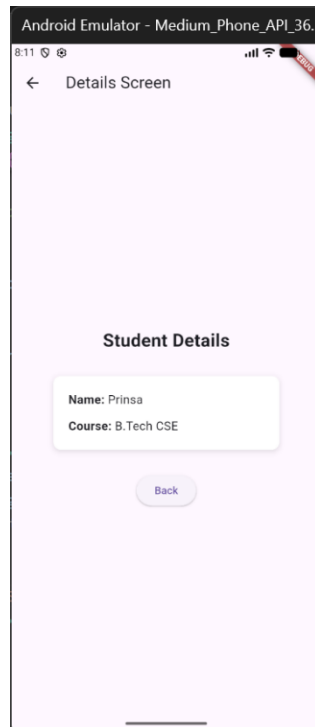
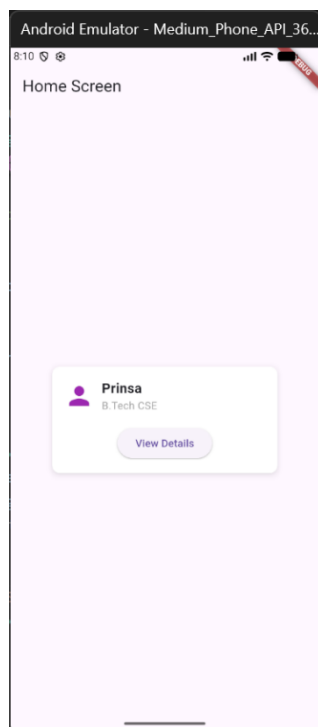
              padding: const EdgeInsets.all(20),
```

```
decoration: BoxDecoration(
  color: Colors.white,
  borderRadius: BorderRadius.circular(10),
  boxShadow: const [
    BoxShadow(
      color: Colors.black12,
      blurRadius: 5,
      offset: Offset(0, 2),
    ),
  ],
),
child: Column(
  crossAxisAlignment: CrossAxisAlignment.start,
  children: [
    // Name Row
    Row(
      children: [
        const Text(
          'Name: ',
          style: TextStyle(
            fontSize: 16,
            fontWeight: FontWeight.bold,
          ),
        ),
```

```
),  
Text(name, style: const TextStyle(fontSize: 16)),  
],  
),  
const SizedBox(height: 12),  
  
// Course Row  
  
Row(  
  children: [  
    const Text(  
      'Course: ',  
      style: TextStyle(  
        fontSize: 16,  
        fontWeight: FontWeight.bold,  
      ),  
    ),  
    Text(course, style: const TextStyle(fontSize: 16)),  
  ],  
),  
],  
),  
),  
const SizedBox(height: 30),  
  
// Back Button
```

```
ElevatedButton(  
  onPressed: () {  
    Navigator.pop(context);  
  },  
  child: const Text('Back'),  
),  
],  
),  
),  
);  
}  
}
```

Output:



5. Explain the concept of data types and write a program to demonstrate type conversion in Dart. Write a Dart program to find the factorial of a number using recursion.

A data type specifies what kind of value a variable can store. It helps Dart allocate memory and perform operations correctly.

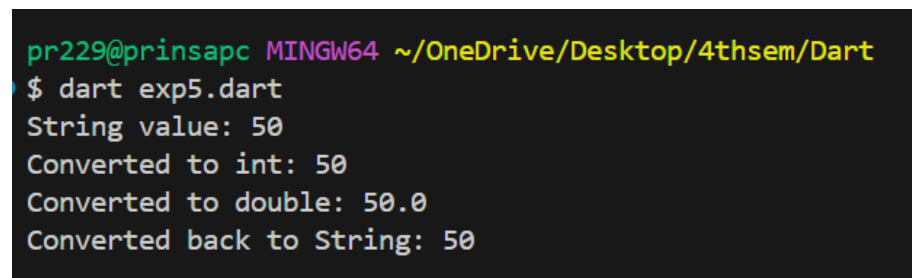
Common Dart Data Types

- int - whole numbers
- double - decimal numbers
- String - text
- bool - true/false
- List - collection of values
- Map - key-value pairs

Program for Type Conversion:

```
void main() {  
    String numStr = "50";  
    int number = int.parse(numStr);  
    double doubleValue = number.toDouble();  
    String backToString = number.toString();  
    print("String value: $numStr");  
    print("Converted to int: $number");  
    print("Converted to double: $doubleValue");  
    print("Converted back to String: $backToString");  
}
```

Output:

A terminal window with a dark background and light-colored text. The prompt is 'pr229@prinsapc MINGW64 ~/OneDrive/Desktop/4thsem/Dart'. The command '\$ dart exp5.dart' has been executed. The output consists of five lines: 'String value: 50', 'Converted to int: 50', 'Converted to double: 50.0', and 'Converted back to String: 50'.

```
pr229@prinsapc MINGW64 ~/OneDrive/Desktop/4thsem/Dart  
$ dart exp5.dart  
String value: 50  
Converted to int: 50  
Converted to double: 50.0  
Converted back to String: 50
```

Program for Factorial Using Recursion:

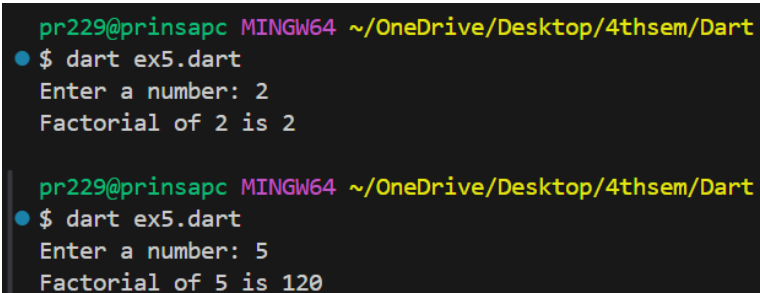
```
import 'dart:io';

int factorial(int n) {
  if (n == 0 || n == 1) {
    return 1;
  } else {
    return n * factorial(n - 1);
  }
}

void main() {
  stdout.write("Enter a number: ");
  int num = int.parse(stdin.readLineSync());

  print("Factorial of $num is ${factorial(num)}");
}
```

Output:



The screenshot shows a terminal window with a dark background. The prompt is 'pr229@prinsapc MINGW64 ~/OneDrive/Desktop/4thsem/Dart'. The user enters '\$ dart ex5.dart'. The program prompts 'Enter a number: 2' and outputs 'Factorial of 2 is 2'. The user enters '\$ dart ex5.dart' again. The program prompts 'Enter a number: 5' and outputs 'Factorial of 5 is 120'.

```
pr229@prinsapc MINGW64 ~/OneDrive/Desktop/4thsem/Dart
● $ dart ex5.dart
Enter a number: 2
Factorial of 2 is 2

pr229@prinsapc MINGW64 ~/OneDrive/Desktop/4thsem/Dart
● $ dart ex5.dart
Enter a number: 5
Factorial of 5 is 120
```

5. What are Material widgets? Use List Tile and Card to create a list of items. Design a responsive layout using `Expanded` and `Flexible` widgets.

Material widgets are Flutter UI components that implement Google's Material Design system. They help create consistent, modern, and responsive interfaces.

Examples:

- AppBar
- Card
- ListTile
- ElevatedButton

```
import 'package:flutter/material.dart';

void main() {

  runApp(const MyApp());

}

class MyApp extends StatelessWidget {

  const MyApp({super.key});

  @override

  Widget build(BuildContext context) {

    return MaterialApp(

      home: Scaffold(

        appBar: AppBar(

          title: const Text("Material Widgets Demo"),

          backgroundColor: Colors.teal,

        ),

        body: Column(

          children: [

            // ----- RESPONSIVE LAYOUT -----
```

```
Row(  
  children: [  
    Expanded(  
      flex: 2,  
      child: Container(  
        height: 60,  
        color: Colors.red,  
        child: const Center(child: Text("Expanded")),  
      ),  
    ),  
    Flexible(  
      flex: 1,  
      child: Container(  
        height: 60,  
        color: Colors.blue,  
        child: const Center(child: Text("Flexible")),  
      ),  
    ),  
  ],  
,  
  const SizedBox(height: 10),  
  const Text(  
    "Item List",
```



```
style: TextStyle(fontSize: 18, fontWeight: FontWeight.bold),
```

```
),
```

```
// ----- CARD + LISTTILE -----
```

```
Expanded(
```

```
child: ListView(
```

```
children: const [
```

```
Card(
```

```
child: ListTile(
```

```
leading: Icon(Icons.book),
```

```
title: Text("Math Book"),
```

```
subtitle: Text("Class 10"),
```

```
),
```

```
),
```

```
Card(
```

```
child: ListTile(
```

```
leading: Icon(Icons.book),
```

```
title: Text("Science Book"),
```

```
subtitle: Text("Class 10"),
```

```
),
```

```
),
```

```
Card(
```

```
child: ListTile(
```

```
leading: Icon(Icons.book),
```

```
title: Text("English Book"),  
subtitle: Text("Class 10"),  
),  
),  
],  
),  
),  
],  
),  
),  
);  
}  
}
```

Output:

